

Carnegie Mellon University — The BXA Intercollege Degrees: Music Performance (Violin) + an Academic Concentration

Program Dossier · Prepared for Eidee Tan & Family · Fall 2027 Entry · Research date: August 15, 2026

Carnegie Mellon is one of very few universities where an advanced violinist can make conservatory-level studio training *half of a single bachelor's degree* while the other half is a serious computing, science, or cognition concentration. That is the BXA Intercollege Degree family: **BCSA** (Computer Science + Arts), **BSA** (Science + Arts), and **BHA** (Humanities + Arts) — all of which can carry a **Music Performance concentration in violin**, with private studio, orchestra, and music theory built into the degree, plus a two-semester capstone that must join both fields. The single most important thing to know: **BXA applicants apply through the School of Music, face a violin prescreen (due December 1) and a live audition, and are reviewed *only* for that College of Fine Arts choice — indicating BXA interest does not guarantee a BXA offer**, and Early Decision is not available for the School of Music. This dossier presents all three BXA variants as a decision family, and notes the non-audition fallback (Cognitive Science B.S. + SHIFT minor with violin as a top extracurricular).

1. The university at a glance

Fact	Detail	Source
Location / type	Pittsburgh, Pennsylvania; private research university of seven schools and colleges	CMU undergraduate admission
Identity line	"My heart is in the work" (Andrew Carnegie) — an arts-and-technology university where five colleges co-sponsor the BXA degrees	BXA Intercollege Degree Programs
Undergraduate enrollment	7,728 (Fall 2025)	Common Data Set 2025–26
Overall admit rate	11.1% (Fall 2025 entry, Class of 2029: 34,867 applied, 3,859 admitted); ED admit rate ≈20.6% (ED not offered for School of Music)	CDS 2025–26
Test policy	Varies by college: School of Computer Science requires SAT/ACT ; Dietrich, Mellon College of Science, Engineering, Tepper are "test-flexible" (a score from SAT/ACT/AP/IB/A-Level is required); College of Fine Arts is test-optional	Standardized testing
SAT middle 50%	1500–1560 composite (EBRW 730–770, Math 770–800); 65.5% of enrolled first-years submitted SAT	CDS 2025–26
Application platform + fee	Common Application exclusively ; \$75 fee , waivers for Common App economic-need indicators	Apply · Admission requirements

Fact	Detail	Source
Deadlines	ED Nov 2 (decision by Dec 15; <i>not available for School of Music</i>); RD Jan 4 — but Drama & Music RD deadline is Dec 1 ; RD decisions by Apr 1	Application plans & deadlines
Financial aid headline	"We meet 100% of demonstrated financial need " (determined by CMU's aid methodology); CSS Profile + FAFSA both required; only U.S. citizens/permanent residents eligible for federal/state aid	Student Financial Services · Prospective students
Interview	Not considered in admission	CDS 2025–26, C7

2. The big picture — what kind of place this is

Carnegie Mellon began as Andrew Carnegie's technical schools and grew into a research university with a peculiar, productive double identity: it is simultaneously one of the world's most respected computing schools and home to a conservatory-style College of Fine Arts that has trained professional musicians, actors, and designers since 1905. Nowhere else in this research set do those two identities share a single campus quadrangle so literally — the School of Music sits a few hundred yards from the School of Computer Science — and the university has institutionalized the connection in the [BXA Intercollege Degree Programs](#), sponsored jointly by five colleges: Fine Arts, Computer Science, Engineering, Dietrich (humanities and social sciences), and Mellon (science).

Undergraduates at CMU are admitted to a specific college, and each college sets its own general-education program. CMU counts coursework in "units" rather than credits (roughly 9–12 units per course; a normal semester is about 45–50 units), and its academic culture is famously work-centered — the unofficial motto really is "my heart is in the work." That culture rewards students who arrive knowing what they want to build, and the BXA degrees are explicitly for students who refuse to choose between an art and an academic discipline.

For a cognition-to-AI-to-law trajectory, CMU's strengths are foundational: it is a birthplace of artificial intelligence and cognitive science (Herbert Simon and Allen Newell did their Nobel- and Turing-recognized work here), the [Center for the Neural Basis of Cognition](#) is a joint CMU–University of Pittsburgh venture dating to 1994, and the [Block Center for Technology and Society](#) studies the economic, organizational, and policy impacts of AI. What CMU does *not* have is a law school; the law-and-policy layer is built from Dietrich's philosophy and policy offerings — most directly the [SHIFT minor](#) described below.

Music at CMU is a professional school, not a campus amenity. Studio faculty include Pittsburgh Symphony Orchestra musicians; instrumental students play in auditioned major ensembles ([Philharmonic](#), [Wind Ensemble](#), [Contemporary Ensemble](#), [Opera Orchestra](#)) as a curricular requirement. The BXA route places a violinist *inside* that professional environment at half-weight — which is exactly its appeal and exactly its cost in schedule and audition risk.

3. The program — BXA with a Music Performance (violin) concentration: BCSA, BSA, and BHA

All BXA degrees share one architecture: a BXA-specific general-education core, one College of Fine Arts concentration, one academic concentration in the partner college, a four-course **BXA seminar sequence**, and a **two-semester senior capstone** that must span both concentrations. The differences are which partner college supplies the academic half. (All unit figures below are from the 2025–26 catalog year pages — the latest published as of August 2026; Fall 2025 concentration sheets.)

The shared music half. The **Music Performance concentration** (Fall 2025 sheet) requires a **minimum of 108 units**: an 85-unit core — Harmony I (9), Solfège I (3), Eurhythmics I (3), Survey of Western Music History (9), Repertoire and Listening for Musicians (1), **four semesters of BXA Studio (36 units)** of private lessons, and **four semesters of Major Ensemble (24 units)** — plus at least 23 units of advisor-approved music electives. **Admission to the performance track requires a live audition (BXA audition & review requirements)**, and for violin a **prescreen video is mandatory first** (details in §8). Music also offers BXA tracks in Composition, Audio Recording & Production, and Sound Theory & Practice, but Performance is the one that matches Eidee’s violin.

The shared connective tissue. The **BXA seminar sequence** is what makes this a designed degree rather than a stapled double major: 52-190 *BXA Seminar I: Building the Wunderkammer* and 52-291 *Transferring Knowledge* (4.5 units each, first-year/sophomore minis) teach interpretive theory and cross-disciplinary method; 52-392 *Deconstructing Disciplines* (9 units, junior spring) prepares the senior project; then the capstone pair — 52-401 *Capstone Project Research* (fall, 9 units) and 52-402 *Capstone Project Production* (spring, 9 units) — produces an original project spanning both concentrations, presented at the university-wide **Meeting of the Minds** research symposium each May. Capstones may be scholarly, creative, experimental, or computational — a music-cognition experiment, an interpretable-AI-for-performance system, a thesis on machine creativity.

The three variants on the table for Eidee

BCSA — Bachelor of Computer Science and Arts (CS + Music). Total **380 units**: 122 gen-ed, a **117-unit School of Computer Science concentration**, the 108–144-unit CFA concentration, 0–33 free electives (**BCSA curriculum**). The **CS concentration** (Fall 2025) is the real SCS core: 15-112 Fundamentals of Programming, then 15-122 Imperative Computation, 15-150 Functional Programming, 15-210 Parallel & Sequential Data Structures and Algorithms, 15-213 Computer Systems, 15-251 Great Ideas in Theoretical Computer Science, 21-127 Concepts of Mathematics, plus **five further SCS courses at 200-level or above** (AI/ML, robotics, HCI, graphics options). CMU describes BCSA students as wanting to “create the tools that make the art they want to make.” This is the strongest technical credential of the three — and the one where cognition, policy, and economics survive only in electives.

BSA — Bachelor of Science and Arts (a Mellon College of Science concentration + Music). Total **380 units**: 129 gen-ed, a **114–145-unit MCS concentration**, the CFA concentration, and **0–29 free electives (BSA curriculum)**. The allowed science concentrations are **Biological Sciences, Chemistry, Environmental & Sustainability Studies, Mathematical Sciences, Neurobiology, and Physics**. Note carefully: **neither Computer Science nor Cognitive Science is a BSA option** — CS lives in BCSA, and Cognitive Science lives in BHA. For Eidee, the only cognition-adjacent BSA concentration is **Neurobiology**, which is a wet-lab/brain-science framing rather than a computational-cognition one. BSA can leave literally zero free electives depending on concentration choice, making it the tightest schedule of the three.

BHA — Bachelor of Humanities and Arts (Cognitive Science + Music), the strong alternative. Total **378 units**: 126 gen-ed, an **81–107-unit Dietrich concentration** chosen from 31 options — **Cognitive Science is explicitly listed** — the CFA concentration, and 1–63 free electives (**BHA curriculum**). The **BHA Cognitive Science concentration** (Fall 2025) is a substantive **87-unit minimum (at least 9 courses)**, not a light minor: 15-112 programming; calculus (21-120 or the 21-111/112 sequence) and 21-127 Concepts of Mathematics; a statistics course (36-218/36-225/36-309/85-309); a three-course computational core — **two of 15-122 Imperative Computation, 15-150 Functional Programming, 15-251 Theoretical CS**, plus one cognitive-modeling course (85-412 Cognitive Modeling, 85-414, 85-419 Neural Computation, or 85-420); a four-course cognitive-psychology core (85-110/85-213, 85-300 research methods, and two advanced electives whose list includes machine learning, perception, logic, and music-cognition options); and one advisor-approved cognitive science elective. This is the variant prior research scored best for Eidee’s cognition-first direction: it keeps roughly two-thirds of the BCSA

computing core *and* adds the psychology of mind, experimental methods, and modeling that the pure CS concentration omits — with the largest free-elective budget (up to 63 units) for law, ethics, and policy courses.

The decision family at a glance

	BCSA: CS + Music	BSA: MCS science + Music	BHA: Cognitive Science + Music
Total units	380	380	378
Academic concentration	CS, 117 units (SCS core + 5 upper-level SCS courses)	114–145 units; options: Bio, Chem, Env. Studies, Math, Neurobiology , Physics — <i>no CS, no CogSci</i>	Cognitive Science, 87 units (programming, math, stats, modeling, cognitive psych)
Free electives	0–33	0–29 (can be zero)	1–63 (largest)
Best when...	Building computational tools (incl. for music/audio) is the core identity	Hard science or brain biology + violin	Human & machine intelligence + violin are the two formal disciplines
Room for law/policy/SHIFT-style courses	Minimal	Minimal	Meaningful (elective budget)
Test policy note	Reviewed via CFA (test-optional), though SCS normally requires tests — submit the 1510 regardless	Reviewed via CFA	Reviewed via CFA
Source	BCSA	BSA	BHA

The non-BXA fallback. If violin should stay outside the degree, the safer CMU route is the **Cognitive Science B.S.** in Dietrich’s Psychology Department (core + pathway + a 27-unit designed concentration, drawing on psychology, linguistics, AI, philosophy, and neuroscience) combined with the **SHIFT minor — formally the “Minor in Societal & Human Impacts of Future Technologies”**, housed in the Philosophy Department: 8 courses / 72 units — two core courses (*AI, Society, and Humanity* and a SHIFT capstone) plus six area courses across Technology, Social & Behavioral Sciences, and Ethics, Policy & Design, with no double-counting limit. SHIFT is the most direct technology-law-and-policy credential CMU offers an undergraduate, and it pairs naturally with Cognitive Science. On this path, violin continues through lessons and ensembles outside the degree (see §6) with no admission audition.

4. How the pieces fit together

A BXA student’s week is genuinely split: studio lesson, ensemble rehearsals (the Philharmonic is a curricular requirement with **seating auditions for string players** each fall, per the **ensembles syllabus**), solfège/harmony classes, and then the CS/CogSci problem-set load on the other side of campus. The architecture is workable because the BXA gen-ed core is trimmed relative to single-college degrees and because the seminar/capstone sequence is shared — but total requirements of 378–380 units against a ~380-unit four-year budget mean the free-elective margin is the slack in the system. BHA preserves the most; BSA can preserve none.

Declaration timing: BXA students declare their specific academic concentration by **spring mid-semester break of sophomore year** (concentration declaration form; **BHA/BSA** pages). A student admitted to BHA can therefore arrive

undecided between, say, Cognitive Science and Behavioral Economics. Changing the *CFA* concentration after arrival requires passing that CFA school’s full admission requirements again.

Gates beyond university admission

- 1. Violin prescreen video — competitive.** Due **December 1** via Acceptd; required of all violin applicants (**violin requirements**).
- 2. Live audition — competitive.** Required for the Music Performance track; unaccompanied; by invitation after prescreen (**BXA audition requirements**).
- 3. BXA offer itself — competitive and separate.** Indicating BXA interest does not guarantee a BXA offer; BXA notification *follows* the initial admission decision (**BXA admission**).
- 4. Academic concentration declaration — requirements-based** (form by sophomore spring).
- 5. Ensemble seating auditions — requirements-based/competitive** each year (placement, not membership).
- 6. Internal transfer into BXA (if not admitted as first-year) — competitive:** requires a completed CMU semester, **cumulative QPA 3.0**, prerequisite coursework, a 500-word Essay of Interdisciplinary Intent, advisor sign-offs, and — critically — **BXA music performance considers internal applicants only in spring for fall enrollment**, with a new audition (**internal transfer**; next deadlines Oct 9, 2026 and Mar 5, 2027).

5. An illustrative four-year path (BHA: Cognitive Science + Music Performance, violin)

Illustrative only — not official advising; rebuild against the Fall 2027 schedule with BXA advisors.

Year	Academic side	Music side	BXA spine
1	15-112 programming; 21-120 calculus; 21-127 Concepts of Math; BHA gen-ed	BXA Studio I-II (violin); Major Ensemble; Harmony I; Solfège I; Eurhythmics I	52-190 Seminar I (spring mini)
2	15-122 + 15-150; statistics; 85-110/85-213 cognitive psychology; declare Cognitive Science concentration by spring break	BXA Studio III-IV; Major Ensemble; Survey of Western Music History; Repertoire & Listening	52-291 Seminar II (spring mini)
3	15-251 or modeling course (e.g., 85-419 Neural Computation); 85-300 Research Methods; advanced cognitive electives (ML, music cognition); begin lab research (CNBC-affiliated labs)	Music electives; ensemble/chamber as scheduled; possible continued studio via electives	52-392 Seminar III (spring)
4	Remaining cognitive electives; free electives spent on SHIFT-style courses (<i>AI, Society, and Humanity</i> ; ethics/policy)	Music electives; performance projects feeding the capstone	52-401 + 52-402 two-semester capstone joining cognition and violin (e.g., expressive timing and predictive models), presented at Meeting of the Minds

For BCSA, replace the academic column with the 15-122 → 15-150 → 15-210 → 15-213 → 15-251 core plus five upper-level SCS courses; for BSA/Neurobiology, with MCS biology/chemistry/neuroscience sequences — and expect near-zero elective slack in both.

6. Beyond the classroom

Cognition and AI research. The [Center for the Neural Basis of Cognition](#) (CMU + University of Pittsburgh, since 1994) anchors Pittsburgh's unusually deep two-university neuroscience ecosystem; BHA Cognitive Science students reach psychology and computational labs through 85-300 methods training and faculty contact, though **no lab placement is ever guaranteed**. BXA's own office highlights student SURF research grants and alumni working "at the intersection of music and AI" ([BXA programs](#)).

Law, policy, ethics. CMU has no law school, but the [Block Center for Technology and Society](#) runs research on responsible AI, the future of work, and technology governance, with undergraduate entry points including a student policy fellowship and a research-assistant resume pool. Curricularly, the [SHIFT minor](#) and its Ethics, Policy & Design area are the most direct classroom complement; a BHA student can approximate SHIFT's content through free electives even if the full minor proves hard to schedule alongside two concentrations.

Violin life. Inside BXA, violin life is the School of Music: weekly private studio (36 units across four semesters), the auditioned-seating [CMU Philharmonic and companion ensembles](#), and performance electives. Outside a music degree (the fallback path), CMU fields an All University Orchestra that performs on the [university events calendar](#); we could not verify current membership/lesson policies for non-majors on official pages, so confirm lesson access with the School of Music (music-admissions@andrew.cmu.edu) before relying on it.

7. Access, feasibility, and honest risks

The application is single-track. This is the structural risk that distinguishes CMU from every non-conservatory path in Eidee's set: a BXA applicant applies through the School of Music and "will not be reviewed for a second college/program outside of your CFA choice" ([BXA admission](#)). There is no simultaneous Dietrich or SCS application. If the audition outcome is negative, the CMU application effectively ends for that cycle. And even a successful School of Music admit is not automatically a BXA admit — the BXA decision arrives separately.

The audition is a conservatory audition. The violin prescreen and live round (memorized Paganini/Kreutzer-class etude, Romantic concerto movement, solo Bach) are judged by a professional string faculty against applicants who intend performance careers. "Advanced" must mean advanced at that standard.

The schedule is real. Two concentrations plus a shared capstone consume essentially the whole 378–380-unit budget. In BSA the free-elective line can literally read zero; in BCSA it is 0–33. Only BHA reliably protects space for the law/policy/ethics layer that motivates Eidee's ten-year plan — which is why prior research ranked BHA Cognitive Science, not the more obvious BCSA, as the best-fit variant.

Law/policy is not in any BXA core. No BXA concentration teaches technology law. That content must come from electives or the SHIFT minor — and whether a BXA student can feasibly add the 8-course SHIFT minor is a scheduling question to put to BXA advising, not an entitlement.

When a different path wins. If violin should remain serious but extracurricular, the **Cognitive Science B.S. + SHIFT minor** in Dietrich is the better CMU application: no audition gate, test-flexible review of a strong academic file, the most direct tech-policy credential on campus, and prior research scored it within a rounding error of the BHA route. If building computational tools (including for music) becomes the core identity, BCSA overtakes BHA. BSA fits only if Eidee's center of gravity moves to laboratory science.

8. Applying — logistics for Fall 2027

CMU uses the **Common Application exclusively** (\$75 fee; waivers via Common App economic-need indicators; essay + three short-answer writing supplement; counselor evaluation + one teacher recommendation) ([requirements](#)). School of Music applicants additionally submit through the **Acceptd portal** (app.getacceptd.com/cmumusic) ([audition & portfolio review](#)).

Item	Date / rule	Source
Early Decision	Not available for the School of Music (ED for other colleges: Nov 2)	Deadlines
Regular Decision — School of Music	December 1 (vs. Jan 4 for other colleges); decisions by April 1	Deadlines
Violin prescreen video	Due December 1 via Acceptd; required of all violin applicants: memorized etude (Kreutzer/Rode/Dont/Fiorillo/Gaviniès or Wieniawski/Paganini caprice), an Allegro movement of a non-Baroque concerto, and a movement of solo Bach	Violin requirements
Live audition	By invitation after prescreen; unaccompanied , same repertoire; in-person in Pittsburgh or NYC regional dates — 2026–27 cycle dates “to be announced” as of research date; monitor the dates page	Audition dates
BXA indication	Select School of Music in the application, then opt into BXA consideration and name the program (BHA/BSA/BCSA); reviewed only for the CFA choice; BXA decision follows the admission offer	First-year process · BXA admission
Testing	CFA is test-optional ; preferred test completion Nov 1 (ED) / Jan 3 (RD); CMU superscores the SAT. With a 1510 superscore, submitting is the obvious choice and keeps every internal-transfer door open (SCS requires tests; Dietrich/MCS are test-flexible)	Standardized testing
Interview	Not offered/considered	CDS 2025–26
Financial aid	CSS Profile + FAFSA both required ; preferred filing Nov 15 (ED) / Feb 15 (RD) ; CMU states it meets 100% of demonstrated need (per its FM+IM methodology); no federal/state aid for non-citizens, limited CMU grants for DACA/select countries	SFS prospective students · Financial aid

Deadlines above are the currently published cycle’s; CMU had not yet posted a distinct 2026–27 admission calendar as of August 15, 2026 — re-verify all dates when the Common App opens.

9. Sources

Admissions: [Application plans & deadlines](#) · [Apply](#) · [Admission requirements](#) · [Standardized testing](#) · [BXA admission page](#) · [Common Data Set 2025–26](#)

Program (BXA): [BXA home](#) · [BHA curriculum](#) · [BCSA curriculum](#) · [BSA curriculum](#) · [BXA seminars & capstone](#) · [BHA Cognitive Science sheet \(Fall 2025\)](#) · [BCSA CS sheet \(Fall 2025\)](#) · [First-year BXA application](#) · [BXA internal transfer](#) · [Cognitive Science B.S.](#) · [SHIFT minor](#)

Music: [BXA Music concentration sheet \(Fall 2025\)](#) · [Violin audition requirements](#) · [BXA audition & review requirements](#) · [Audition & portfolio review](#) · [Audition dates](#) · [Major instrumental ensembles syllabus 2024–25](#)

Aid: [Financial aid overview](#) · [Prospective-student aid steps](#)

Research/centers: [CNBC](#) · [Block Center](#)